STA2002 Probability & Statistics II Lecture 5 - 2023-2024 Term 1

Lecture 5, Two sample test, Power, 'Never confirm the null'

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Suggested Reading: Hogg et al. 2018, Chapter 8.1 - 8.6.

1 Further one–sample test

Suppose data $X_1, \cdots, X_n$ are i.i.d. normal $N(\mu, \sigma^2)$ with unknown μ and σ^2. Suppose we want to test the hypothesis

$$H_0 : \sigma^2 = \sigma_0^2,\ H_1 : \sigma^2 > \sigma_0^2.$$

The test statistics we choose is

$$\widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i,\ S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \widehat{\mu})^2,\ \widehat{T} = \frac{(n-1)\times S^2}{\sigma_0^2},$$

and the rejection region is $[\chi_\alpha, \infty)$. From here, the P–value is

$$\inf\{\alpha \in [0,1] : \widehat{T} \geq \chi_\alpha\} = 1 - \psi(\widehat{T}),$$

where $\psi(\cdot)$ is the cdf of χ^2_{n-1} distribution.

2 Tests comparing different groups

2.1 Two sample tests for means

Recall the CI lecture: $X_{11}, \ldots, X_{1n_1}$ be iid samples from $N(\mu_1, \sigma_1^2)$, and $X_{21}, \ldots, X_{2n_2}$ be iid samples from $N(\mu_2, \sigma_2^2)$. First assume σ_1^2, σ_2^2 are known. Denote the sample mean

$$\bar{X}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} X_{1i}, \quad \bar{X}_2 = \frac{1}{n_2}\sum_{i=1}^{n_2} X_{2i}.$$

Then we have

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} \sim N(0,1)$$

First consider the two-side test

$$H_0 : \mu_1 = \mu_2 \text{ and } \ H_1 : \mu_1 \neq \mu_2$$

Define the test statistic and the rejection region

$$\widehat{T} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} \text{ and } R_\alpha = (-\infty, -z_{\alpha/2}) \cup (z_{\alpha/2}, \infty)$$

then under H_0

$$P\left(\widehat{T} \in R_\alpha\right) = P(\widehat{T} < -z_{\alpha/2}) + P(\widehat{T} > z_{\alpha/2}) = \alpha$$

Under H_1 we have $\widehat{T} = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} + \frac{\mu_1 - \mu_2}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} = N(0,1) + bias$, so for sufficiently large n_1, n_2, $\widehat{T}$ should have large probability to be in R_α.

For calculation of P-value, by the formal definition,

$$\widehat{p} = \inf\{\alpha : \widehat{T} \in R_\alpha\} = \inf\{\alpha : |\widehat{T}| > z_{\alpha/2}\} = \inf\{\alpha : \Phi(|\widehat{T}|) - \Phi(-|\widehat{T}|) > \Phi(z_{\alpha/2}) - \Phi(-z_{\alpha/2}) = 1 - \alpha\}$$
$$\Rightarrow \widehat{p} = 1 - \Phi(|\widehat{T}|) + \Phi(-|\widehat{T}|)$$

Similarly, if we consider one-side alternative

$$H_0 : \mu_1 = \mu_2 \text{ and } \ H_1 : \mu_1 > \mu_2$$

Then the rejection region is constructed as $R_\alpha = (z_\alpha, \infty)$. Then under the null, we have

$$P\left(\widehat{T} \in R_\alpha\right) = P(\widehat{T} > z_\alpha) = \alpha$$

For $\widehat{T} = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} + \frac{\mu_1 - \mu_2}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} = N(0,1) + bias$, under the alternative, we have

$$P\left(\widehat{T} \in R_\alpha\right) = P\left(\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} > z_\alpha - bias\right)$$

and we know that $P\left(\widehat{T} \in R_\alpha\right) \to 1$ as $n \to \infty$. Finally, by definition of $P-$value:

$$\widehat{p} = \inf\{\alpha : \widehat{T} \in R_\alpha\} = \inf\{\alpha : \widehat{T} > z_\alpha\} = \inf\{\alpha : \Phi(\widehat{T}) > 1 - \alpha\}$$
$$\Rightarrow \widehat{p} = 1 - \Phi(\widehat{T})$$

Second, if we assume unknown but equal variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$, from Lecture 3 we have the pooled estimator of σ^2:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

and we have the following test statistics

$$\widehat{T} = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2).$$

1. For two-sided test, the rejection region is $R_\alpha = (-\infty, -t_{\alpha/2, n_1+n_2-2}) \cup (t_{\alpha/2, n_1+n_2-2}, \infty)$, and the $P-$value equals to $\mathbf{P}(|T| \geq |\widehat{T}|)$ where $T \sim t_{n_1+n_2-2}$ and $\widehat{T}$ is the observed value of test statistics.

2. For one-sided test ($\mu_1 > \mu_2$), the rejection region is $R_\alpha = (t_{\alpha, n_1+n_2-2}, \infty)$, and the $P-$value equals to $\mathbf{P}(T \geq \widehat{T})$.

2.2 Two sample tests for proportions

Hypothesis tests comparing two proportions are common in many application areas, such as comparing success rates of a new drug and placebo in a medical clinical trial.

Recall the setting from lecture 3 (Confidence Interval): $X_{11}, \ldots, X_{1n_1}$ be iid samples from $Bernoulli(n_1, p_1)$, and $X_{21}, \ldots, X_{2n_2}$ be iid samples from $Bernoulli(n_2, p_2)$. Denote

$$\hat{p}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} X_{1i}, \quad \hat{p}_2 = \frac{1}{n_2}\sum_{i=1}^{n_2} X_{2i}.$$

Then we have for moderately large sample size n,

$$\frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}2)}{n_2}}} \sim_{Approx} N(0,1)$$

- Consider the two-side test

  $$H_0 : p_1 = p_2 \text{ and } \; H_1 : p_1 \neq p_2$$

  Define the test statistic

  $$\widehat{T} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}2)}{n_2}}}$$

  Then the rejection region is

  $$R_\alpha = (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty)$$

  because under null H_0

  $$P\left(\widehat{T} \in R_\alpha\right) = P(\widehat{T} < -z_{\alpha/2}) + P(\widehat{T} > z_{\alpha/2}) \approx \alpha$$

Now let's find P-value for $\hat{T}$ under H_0

$$\widehat{p} = \inf\{\alpha : \widehat{T} \in R_\alpha\} = \inf\{\alpha : |\widehat{T}| \geq z_{\alpha/2}\} = \inf\{\alpha : \Phi(|\widehat{T}|) - \Phi(-|\widehat{T}|) \geq \Phi(z_{\alpha/2}) - \Phi(-z_{\alpha/2}) = 1-\alpha\}$$
$$\Rightarrow \widehat{p} = 1 - \Phi(|\widehat{T}|) + \Phi(-|\widehat{T}|)$$

- Similarly for one-sided alternative

$$H_0 : p_1 = p_2 \text{ and } \ H_1 : p_1 > p_2$$

The test statistic and rejection region under α are

$$\widehat{T} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}2)}{n_2}}}, \quad R_\alpha = [z_\alpha, \infty)$$

Then

$$\widehat{p} = \inf\{\alpha : \widehat{T} \in R_\alpha\} = \inf\{\alpha : \widehat{T} \geq z_\alpha\} = \inf\{\alpha : \Phi(\widehat{T}) \geq \Phi(z_\alpha) = 1-\alpha\}$$
$$\Rightarrow \widehat{p} = 1 - \Phi(\widehat{T})$$

- For one-sided alternative

$$H_0 : p_1 = p_2 \text{ and } \ H_1 : p_1 < p_2$$

The test statistic and rejection region under α are

$$\widehat{T} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}2)}{n_2}}}, \quad R_\alpha = (-\infty, -z_\alpha]$$

Then

$$\widehat{p} = \inf\{\alpha : \widehat{T} \in R_\alpha\} = \inf\{\alpha : \widehat{T} \leq -z_\alpha\} = \inf\{\alpha : \Phi(\widehat{T}) \leq \Phi(-z_\alpha) = \alpha\}$$
$$\Rightarrow \widehat{p} = \Phi(\widehat{T})$$

3 Take care of the type-II error, 'never confirm the null'

So far we have introduced the procedure of a test. When you explain the test result to other people, you may simply say like 'the test prefer H_0', 'the test reject H_0', etc. However, as a statistician, you should be aware of the situation that the test does not reject H_0. To see why, let us consider the following 3 examples:

Suppose $X_1, \cdots, X_n$ are i.i.d. normal random variables with unknown μ and known σ. We want to test

$$H_0 : \mu = \mu_0 \text{ versus } H_1 : \mu > \mu_0$$

Consider three cases: 1. $\mu = \mu_0 + \delta$, 2. $\mu = \mu_0 + \delta/\sqrt{n}$, 3. $\mu = \mu_0 + \delta/n$. Here δ is a constant. We use the test statistics $\widehat{T} = \frac{\sqrt{n}(\widehat{\mu}-\mu_0)}{\sigma}$ and the rejection region $R_\alpha = (z_\alpha, \infty)$. Suppose $\widehat{T} \notin R_\alpha$, which means that we do not reject the null hypothesis. Let us see what happens:

Case 1. In that situation $\widehat{T} = \frac{\sqrt{n}(\widehat{\mu}-\mu)}{\sigma} + \frac{\sqrt{n}\delta}{\sigma}$, so we have the bias tends to ∞ as $n \to \infty$, so as the sample size increases, it is not likely for $\widehat{T} \notin R_\alpha$ to happen.

Case 2. In that situation $\widehat{T} = \frac{\sqrt{n}(\widehat{\mu}-\mu)}{\sigma} + \frac{\delta}{\sigma}$, and

$$P(\widehat{T} \notin R_\alpha) = P(\frac{\sqrt{n}(\widehat{\mu}-\mu)}{\sigma} \leq z_\alpha - \frac{\delta}{\sigma})$$

i.e., despite the possibility of $\widehat{T} \notin R_\alpha$ decreases due to the bias, but the shrink is limited. E.g., if $\delta/\sigma = 0.4$, notice that $z_{95\%} \approx 1.7$ and $z_{90\%} \approx 1.3$, so we still have a chance of 90% for not rejecting H_0.

Case 3. In that situation $\widehat{T} = \frac{\sqrt{n}(\widehat{\mu}-\mu)}{\sigma} + \frac{\delta}{\sqrt{n}\sigma}$, so we have

$$P(\widehat{T} \notin R_\alpha) = P(\frac{\sqrt{n}(\widehat{\mu}-\mu)}{\sigma} \leq z_\alpha - \frac{\delta}{\sqrt{n}\sigma})$$

i.e., the test even cannot observe the difference between μ and μ_0 asymptotically.

In this example, we can see that, despite we are confident to say a test rejects H_0(with some probability), strictly speaking, we cannot say a test confirms H_0 due to the marginal situation. So, a more precise description is 'we do not have evidence to reject H_0'.

4 Power and Sample Size

Recall the definition of Type I and Type II Error:

1. The probability of rejecting the null hypothesis H_0 when it is true is defined as the Type-I error.
2. The probability of failing to reject the null hypothesis H_0 when it is false is defined as the Type-II error.

4.1 Power

Definition 4.1. *The* **Power function** $K(\theta)$ *of a hypothesis test with rejection region* R *is* $\mathbf{P}_\theta(X \in R)$.

Suppose $\beta(\theta)$ is the Type-II error, then we have $K(\theta) = 1 - \beta(\theta)$.

Power for testing normal mean Assume $X_1, \ldots, X_n$ iid $N(\mu, \sigma^2)$ with known σ^2. First consider the two sided hypothesis testing

$$H_0 : \mu = \mu_0 \quad vs. \quad H_1 : \mu \neq \mu_0.$$

For a given rejection region $R := \{\sqrt{n}|\bar{x} - \mu_0|/\sigma > z_{\alpha/2}\}$. The power function under $\mu \neq \mu_0$ is

$$\begin{aligned} K(\mu) &= \mathbf{P}_\theta(\frac{\sqrt{n}|\bar{x} - \mu_0|}{\sigma} > z_{\alpha/2}) \\ &= 1 - \mathbf{P}_\theta(\frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} - z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{x} - \mu)}{\sigma} \leq \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_{\alpha/2}) \\ &= 1 - \Phi(\frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_{\alpha/2}) + \Phi(\frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} - z_{\alpha/2}) \\ &= \Phi(\frac{\sqrt{n}(\mu - \mu_0)}{\sigma} - z_{\alpha/2}) + \Phi(-\frac{\sqrt{n}(\mu - \mu_0)}{\sigma} - z_{\alpha/2}) \end{aligned}$$

note that the power function $K(\mu) \to 1$ when $n \to \infty$ or $|\mu| \to \infty$, which means in such cases the null hypothesis can be correctly rejected with probability close to one. Moreover, the power function $K(\mu)$ is minimized at $\mu = \mu_0$ (Why?).

Similarly, for one-sided test

$$H_0 : \mu = \mu_0 \quad vs. \quad H_1 : \mu > \mu_0.$$

The rejection region is $R := \{\sqrt{n}(\bar{x} - \mu_0)/\sigma > z_\alpha\}$, and the power function

$$\begin{aligned} K(\mu) &= \mathbf{P}_\theta(\sqrt{n}(\bar{x} - \mu_0)/\sigma > z_\alpha) = 1 - \mathbf{P}_\theta(\frac{\sqrt{n}(\bar{x} - \mu)}{\sigma} \leq \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha) \\ &= 1 - \Phi(\frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha) = \Phi(\frac{\sqrt{n}(\mu - \mu_0)}{\sigma} - z_\alpha) \end{aligned}$$

and $K(\mu) \to 1$ when $n \to \infty$ or $\mu \to \infty$, and $\beta(\mu) \leq \alpha$ for $\mu \leq \mu_0$.

We left the calculation of the power function for

$$H_0 : \mu = \mu_0 \quad vs. \quad H_1 : \mu < \mu_0.$$

as an exercise. Moreover, try plotting the power function by yourself.

Comments From the above calculation of the power function for normal examples, we can see several factors affecting the power:

1. If the significance level α decreases, the power will decrease;
2. If the alternative mean is shifted farther away from the null mean, the power increases;
3. If the standard deviation of the distribution of individual observations increases, the power decreases;

4. If the sample size n increases, the power increases.

Power for two samples $X_{11}, \ldots, X_{1n_1}$ be iid samples from $N(\mu_1, \sigma_1^2)$, and $X_{21}, \ldots, X_{2n_2}$ be iid samples from $N(\mu_2, \sigma_2^2)$, with known variances. Consider the test

$$H_0 : \mu_1 = \mu_2 \quad vs. \quad H_1 : \mu_1 \neq \mu_2.$$

Suppose the alternative difference is $\Delta = \mu_1 - \mu_2$. The power function

$$\begin{aligned}
K(\Delta) &= \mathbf{P}(|\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}}| \geq z_{\alpha/2}) \\
&= 1 - \mathbf{P}(\frac{-\Delta}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} - z_{\alpha/2} \leq \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} \leq \frac{-\Delta}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} + z_{\alpha/2} \\
&= 1 - \Phi(\frac{-\Delta}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} + z_{\alpha/2}) + \Phi(\frac{-\Delta}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} - z_{\alpha/2}) \\
&= \Phi(\frac{\Delta}{\sqrt{(\frac{\sigma_1^2}{n_1} - \frac{\sigma_2^2}{n_2})}} + z_{\alpha/2}) + \Phi(\frac{-\Delta}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} - z_{\alpha/2})
\end{aligned}$$

In particular, if we have $n_1 = n_2 = n$ and $\sigma_1^2 = \sigma_2^2 = \sigma^2$, then we have

$$K(\Delta) = \Phi(\frac{\Delta}{\sigma}\sqrt{\frac{2}{n}} - z_{\alpha/2}) + z_{\alpha/2}) + \Phi(-\frac{\Delta}{\sigma}\sqrt{\frac{2}{n}} - z_{\alpha/2}).$$

What is the power function for one-sided tests?

4.2 Sample Size

First consider the one-sided test

$$H_0 : \mu = \mu_0 \quad vs. \quad H_1 : \mu > \mu_0.$$

Suppose the true alternative mean value is $\mu_1 > \mu_0$. What sample size is needed to be able to detect a significant difference (reject H_0) with probability at least $1 - \beta$?

$$K(\mu_1) = \Phi(\frac{\sqrt{n}(\mu_1 - \mu_0)}{\sigma} - z_\alpha) \geq 1 - \beta \Rightarrow \frac{\sqrt{n}(\mu_1 - \mu_0)}{\sigma} - z_\alpha \geq z_\beta \Rightarrow n \geq \frac{(z_\beta + z_\alpha)^2\sigma^2}{(\mu_1 - \mu_0)^2}$$

which means the smallest sample size needed is $\frac{(z_\beta + z_\alpha)^2\sigma^2}{(\mu_1 - \mu_0)^2}$.

For two-sided tests, suppose the true alternative mean value is $\mu_1 > \mu_0$, if we want the alternative

can be detected with power $1-\beta$, we need

$$K(\mu_1) = \Phi(\frac{\sqrt{n}(\mu_1-\mu_0)}{\sigma} - z_{\alpha/2}) + \Phi(-\frac{\sqrt{n}(\mu_1-\mu_0)}{\sigma} - z_{\alpha/2}) = 1-\beta$$

since $\mu_1 > \mu_0$, we have

$$\Rightarrow \Phi(\frac{\sqrt{n}(\mu_1-\mu_0)}{\sigma} - z_{\alpha/2}) \approx 1-\beta \Rightarrow n \approx \frac{(z_\beta + z_{\alpha/2})^2\sigma^2}{(\mu_1-\mu_0)^2}$$

References

Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman (2018). *Probability and statistical inference.*